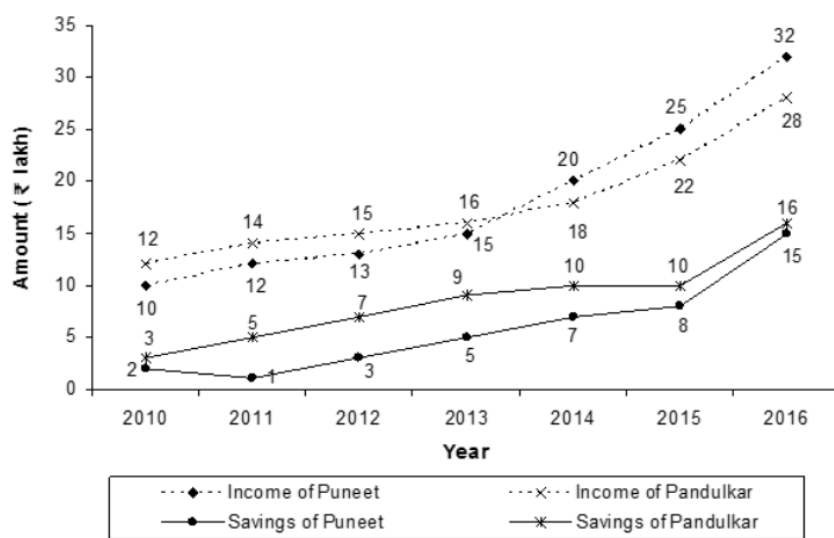


DI LINE GRAPHS 13-9-26

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

The line graph gives the annual income and savings of two persons – Puneet and Pandulkar, across seven years.



In which of the following years did Puneet had higher income than Pandulkar for the first time?

In which of the following years did Puneet had higher income than Pandulkar for the first time?

- ☐ a) 2012
- ☐ b) 2013
- ☐ c) 2014
- ☐ d) 2015

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Solution:

On observation, we can say that in the year 2014, the income of Puneet is more than that of Pandulkar for the first time.

Choice (C)

If the difference between the incomes of a person in a year and its previous year represents the hike that person got in the year, then in which year did Pandulkar get the maximum hike?

- ☐ a) 2012
- ☐ b) 2013
- ☐ c) 2014
- ☐ d) 2016

Solution:

$$\begin{aligned}\text{Hike that Pandulkar got in 2012} &= 15 - 14 = 1 \\ 2013 &= 16 - 15 = 1 \\ 2014 &= 18 - 16 = 2 \\ 2016 &= 28 - 22 = 6\end{aligned}$$

Choice (D)

Among the following years, in which year was the expenditure of Puneet the highest?

- ☐ a) 2011
- ☐ b) 2012
- ☐ c) 2013
- ☐ d) 2016

Solution:

$$\begin{aligned}\text{Expenditure of Puneet in 2011} &= 12 - 1 = 11 \\ 2012 &= 13 - 3 = 10 \\ 2013 &= 15 - 5 = 10 \\ 2016 &= 32 - 15 = 17\end{aligned}$$

Choice (D)

If savings percentage = $\frac{\text{savings}}{\text{income}} \times 100$, then what is the savings percentage of Pandulkar in the year 2015?

- ☐ a) $45\frac{5}{11}\%$
- ☐ b) $46\frac{7}{8}\%$
- ☐ c) 50%
- ☐ d) 46%

Solution:

$$\text{Savings percentage of Pandulkar in 2015} = \frac{10}{22} \times 100 = 45\frac{5}{11}\%$$

Choice (A)

What is the difference (in Rs. lakh) in the expenditures of Puneet and Pandulkar in the year 2013?

- ☐ a) 4
- ☐ b) 3
- ☐ c) 5
- ☐ d) 6

Solution:

$$\begin{aligned}\text{Expenditure of Puneet in 2013} &= 15 - 5 = 10 \\ \text{Expenditure of Pandulkar in 2013} &= 16 - 9 = 7 \\ \therefore \text{Difference in their expenditure} &= 10 - 7 = 3\end{aligned}$$

Choice (B)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The following table gives the information of the production, storage and loss due to storage of XYZ Ltd.

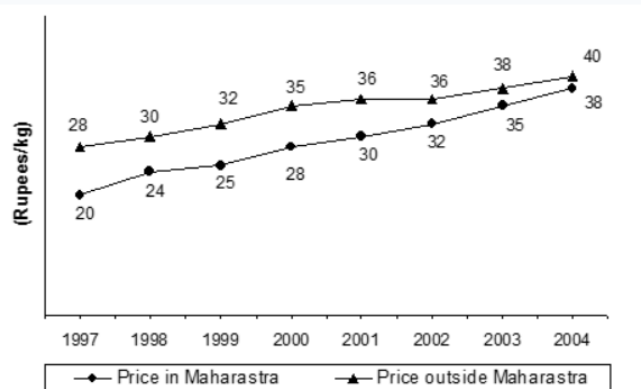
Year	Production (Thousand tonnes)	Storage (Thousand tonnes)	Loss due to storage (Thousand tonnes)
1997	80	42	7
1998	95	45	5
1999	98	50	2
2000	110	50	4
2001	113	56	7
2002	121	62	6.5
2003	124	68	8.5
2004	130	69	9

Quantity sold in Maharashtra = Production – Storage

Quantity sold outside Maharashtra = Storage – Loss due to storage

Note: XYZ Ltd. sells the product in Maharashtra and outside Maharashtra.

The selling price per kg of the product in Maharashtra and outside Maharashtra for 8 years is given in the graph below.



Directions for questions 6 and 7: Type in your answer in the input box provided below the question.

What is the percentage increase in the total revenue of XYZ Ltd. from 1998 to 1999?

Solution:

The revenue in 1998

$$= [24 \times 50 + 30 \times (45 - 5)] \times 10^8 = 240 \times 10^7$$

The revenue in 1999

$$= [25 \times 48 + 32 \times (50 - 2)] \times 10^8 = 273.6 \times 10^7$$

$$\text{The required percentage} = \frac{273.6 - 240}{240} \times 100 = 14\%$$

Ans : (14)

Directions for questions 6 and 7: Type in your answer in the input box provided below the question.

In which year was the quantity sold outside Maharashtra, as a percentage of the total production in that year, the highest?

Solution:

Year	Required Percentage
------	---------------------

1997	$\frac{42-7}{80} \times 100 = \frac{35}{80} \times 100 = 43.75$
------	---

1998	$\frac{40}{95} \times 100 = 42.10$
------	------------------------------------

1999	$\frac{48}{98} \times 100 = 48.97$
------	------------------------------------

2000	$\frac{46}{110} \times 100 = 41.81$
------	-------------------------------------

2001	$\frac{49}{113} \times 100 = 43.36$
------	-------------------------------------

2002	$\frac{55.5}{121} \times 100 = 45.86$
------	---------------------------------------

2003	$\frac{59.5}{124} \times 100 = 47.98$
------	---------------------------------------

2004	$\frac{60}{130} \times 100 = 46.15$
------	-------------------------------------

The highest was in 1999.

Ans : (1999)

Directions for question 8: Select the correct alternative from the given choices.

In which of the following years was the revenue earned the highest?

- ☐ a) 2004
- ☐ b) 2002
- ☐ c) 2003
- ☐ d) 2000

Solution:

Revenue earned in 2000 = $60 \times 28 + 46 \times 35 = 3290$
Similarly revenues earned in 2002, 2003, and 2004 are 3886, 4221 and 4718.
 $\therefore$ It has the highest in 2004.

Alternate solution:

As the production and the sales outside Maharashtra are both highest in 2004 along with the individual prices, the revenue is the highest in 2004.

Choice (A)

Directions for questions 9 and 10: Type in your answer in the input box provided below the question.

In how many years from 1998 to 2003 was the loss due to storage less in that year, as compared to its previous year as well as its next year?

Your Answer: Unattempted

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Solution:

In 1999 and 2002.

Ans : (2)

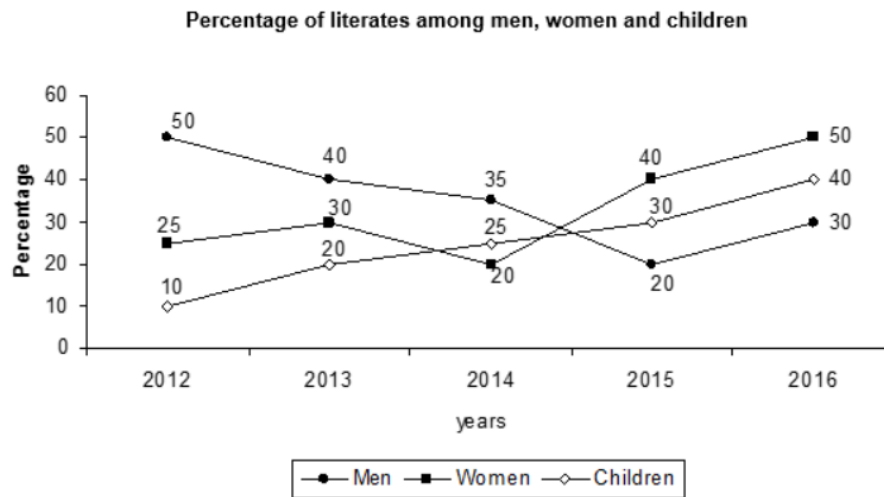
Directions for questions 9 and 10: Type in your answer in the input box provided below the question.

In how many years, was the loss due to storage more than ten percent of the storage?

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The table shows the total number of men, women and children over the years.

Years	Total population	Percentage of the total population		
		Men	Women	Children
2012	1,00,000	20	30	50
2013	1,20,000	30	25	45
2014	1,25,000	24	28	48
2015	1,40,000	35	35	30
2016	1,50,000	25	35	40



The number of literate women in 2012 is what percent of the total number of women in 2013?

- ☐ a) 20%
- ☐ b) 25%
- ☐ c) $71\frac{3}{7}\%$
- ☐ d) 75%

Solution:

Number of literate women in 2012

$$= 100000 \times \frac{30}{100} \times \frac{25}{100} = 7500$$

$$\text{Number of women in 2013} = 120000 \times \frac{25}{100} = 30000$$

$$\text{Required percentage} = \frac{7500}{30000} \times 100$$

$$= 25\%$$

Choice (B)

What is the ratio of number of illiterate men to the number of literate children in the year add 6?

- ☐ a) 35 : 8
- ☐ b) 32 : 35
- ☐ c) 35 : 32
- ☐ d) 35 : 4

Solution:

$$\text{Required ratio} = 25 \times 70 : 40 \times 40 = 35 : 32$$

Choice (C)

What is the total number illiterate children in the years 2012 and 2013 together?

- ☐ a) 88200
- ☐ b) 89200
- ☐ c) 89700
- ☐ d) 88700

Solution:

Number of illiterate children in 2012 and 2013

$$= 10000 \times \frac{50}{100} \times \frac{90}{100} + 120000 \times \frac{45}{100} \times \frac{80}{100}$$
$$= 45000 + 43200 = 88200$$

Choice (A)

What is average number of men over the given period?

- ☐ a) 33500
- ☐ b) 34500
- ☐ c) 37500
- ☐ d) 38500

Solution:

$$\text{Required average} = \frac{1000 \times 20 + 1200 \times 30 + 1250 \times 24 + 1400 \times 35 + 1500 \times 25}{5}$$
$$= 34500$$

Choice (B)

The number of men who are literates in 2013 is what percent of the number of women who are illiterates in 2014?

- ☐ a) $14\frac{2}{7}\%$
- ☐ b) $71\frac{3}{7}\%$
- ☐ c) $51\frac{3}{7}\%$
- ☐ d) $42\frac{6}{7}\%$

Solution:

Number of literate men in 2013

$$= 120000 \times \frac{30}{100} \times \frac{40}{100} = 14400$$

Number of illiterate women in 2014

$$= 125000 \times \frac{28}{100} \times \frac{80}{100} = 28000$$

$$\text{Required percentage} = \frac{14400}{28000} \times 100 = 51\frac{3}{7}\%$$

Choice (C)